

James DeSilver

I build systems that *model or perceive* the real world — food-access simulations, competition robots, and one rewired arcade cabinet.

jdesilver@wisc.edu

LinkedIn

GitHub

STUDYING

University of Wisconsin–Madison

B.S. Computer Science & Data Science, Mathematics minor · Sep 2025 – May 2029

- Letters & Science Honors Program.
- Undergraduate Research Scholar, 2025–26 — a mentored placement pairing early undergraduates with faculty-led research.
- Coursework in object-oriented programming, operating systems, data science, and machine learning.

CURRENT WORK

Software Engineer — Kaufman Lab, FEAST

UW-Madison · Sep 2025 – present

- Build features for FEAST (Food Equity Access Simulation Technology), an agent-based model of how store openings and closures change a community's access to food.
- Work across the stack with undergraduate, graduate, and postdoctoral researchers on a branch-and-pull-request Git workflow.
- Rebuilt simulation controls and distance-modeling components, and evaluated optimization algorithms to cut simulation runtime.
- Ran market research in R that shaped data strategy for 50+ local food systems.

Secretary — Kappa Eta Kappa, Delta Chapter

Madison, WI · Apr 2026 – present

- Keep the chapter record: Executive Board and general meeting minutes, attendance, excusals, and active/inactive/alumni rosters.
- Rebuilt check-in and document templates so meetings run without administrative drag
- Previously Executive Board Member-at-Large (Dec 2025 – Apr 2026), carrying member feedback to the board and leading committees to deadline.

Advising Fellow — Matriculate

Remote · Dec 2025 – present

- Near-peer college advisor for high-achieving, low-income high school students, from building a college list through final enrollment.
- Advise three fellows on essays, supplements, FAFSA and CSS Profile, and admissions decisions; completed 100% of milestones on or ahead of deadline.

Rated Exceeding Expectations – Matriculate's top performance tier

THINGS I'VE BUILT

Arcade cabinet and retro strategy game

Kappa Eta Kappa · Engineering EXPO · Jan – Apr 2026

- Co-built a retro strategy game in Godot in the vein of Clash Royale, writing the core loop and game logic from scratch.
- Rewired and reprogrammed a six-year-old arcade cabinet around an NVIDIA Jetson, retro displays, and original controllers.
- Worked with a multidisciplinary engineering team on gameplay systems, debugging, and hardware integration, coordinating through GitHub.

2nd place – UW-Madison Engineering EXPO, 2026

Competition robotics — FIRST Tech Challenge

Software engineering lead · Sep 2023 – May 2025

- Led ten programmers and engineers through statewide championships to a worldwide finish.
- Designed and programmed competition robots in Java under live competition constraints.
- Integrated convolutional neural networks with OpenCV and TensorFlow for on-robot vision, reaching 99% detection accuracy.
- Raised \$3K+ in sponsorship and presented the work to judge panels and audiences of 1,000+.

4th worldwide of 7,600 teams – FTC World Championship, 2025

Text-based RPG in the Elden Ring universe

Personal project · Apr 2023 - Jan 2024

- Architected the game in Java around custom classes, inheritance, and polymorphism to model characters, enemies, items, and locations.
- Implemented a leveling and stat-progression loop, combat, and branching exploration, then balanced it through self-directed testing and refactoring.

EARLIER WORK

Dockhand, Levels 1–2 — Bridge Marina

Lake Hopatcong, NJ · Jun 2025 - Aug 2026

Docked, launched, and secured boats; kept the fleet and docks ready and compliant; trained new hires on safe on-water procedure.

Artisan — Jersey Artisan Co.

Morris Plains, NJ · Summer 2026

Ran booth operations at three recurring farmers markets end to end, and forecast per-event quantities from market-specific demand to cut both overstock and stockouts.

Crew Member — Main Lake Market

Lake Hopatcong, NJ · Summer 2026

Balanced kitchen, counter, and marina fuel dock at once through peak hours in a high-volume seasonal shop.

Counselor — Willow Lake Day Camp

Jefferson, NJ · Jun 2022 - Aug 2024

Led and supervised camper groups through daily programming; promoted from counselor-in-training to junior counselor to counselor.

TOOLKIT

LANGUAGES

Python, C++, Java, JavaScript, R, HTML, CSS

MACHINE LEARNING AND VISION

TensorFlow, OpenCV, convolutional neural networks, agent-based modeling, LLM APIs

ENGINES AND PLATFORMS

Godot, Unreal Engine, NVIDIA Jetson, Microsoft Azure

TOOLS

Git, GitHub, VS Code, Android Studio

SPOKEN

English (native), Spanish